

[Opening — Title Slide]

Good morning/afternoon, respected committee members. I am Anoop V. Eluvathingal, a PhD candidate in the Department of Electrical Engineering at IIT Madras, working under the supervision of Prof. K. Shanti Swarup.

Today I present my thesis titled **“Advanced Protection for Distributed Power Systems and Microgrids”** for the Thesis Submission Approval.

My work addresses a fundamental challenge in modern power grids: how do we protect networks that are increasingly dominated by inverter-based resources, which behave fundamentally differently from the synchronous generators our relay algorithms were designed for?

Over the next 20 minutes I will walk you through the research background, the six key contributions, and our validated results. Backup slides covering each thesis chapter in detail are available after the last numbered slide for the Q&A.

└ Outline

- Background and Context
- Research Motivation
- State-of-the-Art & Limitations
- Research Objectives
- Key Contributions (1 of 3)
- Key Contributions (2 of 3)
- Key Contributions (3 of 3)
- Thesis Organization
- List of Publications (1 of 2)
- List of Publications (2 of 2)
- Summary & Next Steps

[Slide 1 — Outline]

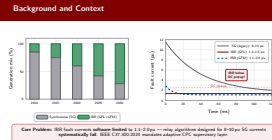
I have organised this presentation into 12 sections.

We begin with the **background and motivation**, then critically examine what the **state of the art** can and cannot do.

The core of the talk covers **six key contributions** spread across three slides — one per two contributions — followed by the **thesis organisation, publications**, and a closing **summary with recommended next steps**.

Please hold questions until the end; I am happy to go to any backup chapter slide during discussion.

Background and Context



[Slide 2 — Background and Context]

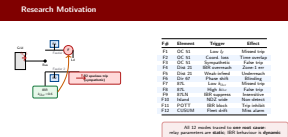
The left chart shows the generation mix transformation. By 2030, IBRs are projected to supply 72% of generation while synchronous machines drop to 28%. This is already happening in India's renewable-rich states.

The right chart shows the **critical technical consequence**. A synchronous generator produces 8–10 pu fault current. An inverter-based resource is software-limited to 1.1–2.0 pu.

The red dotted line is a typical overcurrent relay pickup set for synchronous machines. When an IBR feeds a fault, the current simply does not reach the pickup threshold — the relay does not see a fault at all.

This is the physical root of every failure mode I will show next. IEEE C37.300-2024, the new CPC systems standard, explicitly mandates an adaptive supervisory layer for exactly this reason.

Research Motivation



[Slide 3 — Research Motivation]

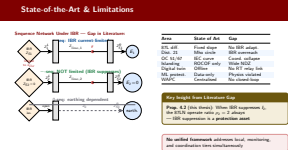
The 12 failure modes on the right are not theoretical. They are systematically documented from grid incident reports and protection literature.

The diagram on the left illustrates the most **insidious mode** — **sympathetic tripping**. A fault on Feeder 1 causes IBR on Feeder 2 to push current into the fault through the common bus. R2 on the healthy feeder sees this elevated current and trips unnecessarily, disconnecting healthy load.

The table spans all major protection elements: overcurrent 51, distance 21, directional 67, line differential 87L, islanding, pilot schemes, and CUSUM-based monitoring.

All 12 modes trace to one root cause: relay parameters are static while IBR behaviour is dynamic. The conclusion is clear — we need an adaptive supervisory layer that continuously refreshes relay settings from real-time estimates.

└ State-of-the-Art & Limitations



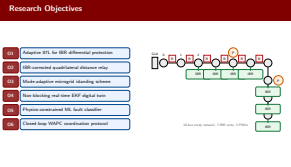
[Slide 4 — State-of-the-Art & Limitations]

The sequence network diagram reveals a critical gap. In the positive-sequence network, IBR current is limited to k_{ibr} . But in the **negative-sequence network**, the IBR presents zero impedance — it does not limit negative-sequence current.

This leads to Proposition 4.2 of the thesis: when IBR suppresses l_2 , the 87LN operate ratio $\rho_2 = 2$ always, regardless of k_{ibr} . IBR suppression is therefore a **protection asset** for negative-sequence elements — an insight not previously reported.

The gap table summarises seven protection areas. Each row represents a specific chapter of this thesis. The final row highlights the absence of any closed-loop, multi-tier coordination framework — which is exactly what SAMBP provides.

Research Objectives



[Slide 5 — Research Objectives]

The six objectives map directly to the six contributions. O1 through O3 address local relay adaptation: line differential, distance, and microgrid islanding. O4 and O5 address the monitoring layer: real-time parameter estimation and physics-informed machine learning. O6 closes the loop at the coordination layer.

The study network on the right is the common testbed for all six contributions. It has 10 buses, 7 IBR units, and 3 PMUs, configured to represent a realistic distribution-level active network. All Monte Carlo campaigns were run on this network.

The k_{ibr} parameter — defined as the IBR current limit normalised by rated current — ranges from 0.03 to 0.85 pu across the study scenarios, covering everything from near-zero to near-rated IBR output.

Key Contributions (1 of 3)

C1 — Adaptive 87L, Zero Differential
 First reported k_m adaptive slope law
 $k_m^* = \frac{0.5(1 - k_{ibr})^2}{(1 + k_{ibr})^2} \cdot \frac{0.025k_{ibr}}{1 + k_{ibr}} + 0.05$
 + 87LN zero sequence direction (pure-IBR network fault)
 + TDCS transient differential reach (non- k_{ibr} blind spot)
 Results: TPR = 1.000 for all bus < 30.00, 0.995 for 30.01 through 30.20, 0.99999 for 30.21 through 30.40000 MVA reach only

C2 — IBR Command Distance Relay
 Directional reach correction:
 $k_m^{correct} = k_m(1 + 0.01k_{ibr})$
 Quadrilateral reach regions only, < 7.6 pp detection gain
 (30.21 through 30.40, 10000 MC trials)

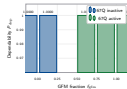


FIG. 10. SAMPF detects all faults for any 87Q/GFL mix

[Slide 6 — Key Contributions 1 of 3]

C1 — Adaptive 87L: The core result is the adaptive slope law $k_m^*(k_{ibr})$. A fixed slope fails at both extremes of k_{ibr} : too sensitive at high k_{ibr} causing false trips; too insensitive at low k_{ibr} causing missed trips. Our law tracks the through-fault operate ratio in real time. We also introduce 87LN for zero-sequence sensitivity in pure-IBR networks, and TDCS — Transient Differential Catch Scheme — as a backstop for the sub-threshold blind spot.

The bar chart confirms 100% detection for any GFL/GFM mix. The 67Q element activates only above 50% GFM fraction when negative-sequence directional discrimination becomes meaningful.

C2 — Adaptive distance: IBR phase angle injection shifts the apparent impedance outside the relay mho circle. The adaptive quadrilateral reach corrects for this with a +7.6 pp detection gain validated over 10 000 Monte Carlo trials.

Key Contributions (2 of 3)



[Slide 7 — Key Contributions 2 of 3 — SAMBP Framework]

This diagram is the architectural summary of the entire thesis. It is important to read this correctly.

Tier 1 (blue, left) is the local relay layer — adaptive 87L, sequence elements, and IBR modelling. This is Chapters 2–4.

Tier 2 (lighter blue) extends to distance and microgrid protection. Chapters 5–6.

Tiers 3 and 4 (green) form the monitoring layer — EKF digital twin and PINN classifier. Chapter 7 and 8.

Tier 5 (maroon) is the Wide-Area Protection Coordinator which sends updated settings back to the relays. Chapter 9.

Tier 6 (gold) is the Digital Twin validation and audit framework spanning Chapters 7–9.

The arrows between tiers show the upward information flow. This is a **closed feedback loop**, not a hierarchical command system. The relay can always act on its own physics; the upper tiers only refine it.

Key Contributions (3 of 3)

[Slide 8 — Key Contributions 3 of 3]

C3 — Islanding detection: The combined ROCOF plus V_2/V_1 criterion closes the non-detection zone to below 1% of the load range, while maintaining IEEE 1547-2018 ride-through compliance. Setting-group switching via GOOSE completes in 50 ms.

C4 — EKF digital twin: The dual-core shadow-register design ensures the TRIP core always uses the most recent committed estimate. The EKF estimation cycle of 20 ms is completely decoupled from the relay's 50 μ s trip decision.

C5 — PINN: Physics constraints — KVL, $I_1 = I_2^*$ symmetry, and current limiting — are embedded directly in the loss function. At 96.1% 5-class accuracy with zero physics violations and 3.7 μ s inference, this is directly deployable in real relay firmware.

C6 — WAPC: The closed-loop protocol runs CUSUM monitoring, parameter computation, digital twin validation, MMS upload, and relay activation in a total of 31 minutes. HMAC-SHA256 authentication reduces GOOSE message spoofing risk by 54.9%.

C3 — Mode Adaptive Islanding Detection
 Combined criterion: $|dV/dt| > 0.5 \text{ Hz/s}$ AND $|dV/V| > 2\%$
 OR $V_2/V_1 > 0.82$
 IEEE 1547-2018 compliant, NZSC < 1% load range
 Setting-group switch via GOOSE in 50 ms

C4 — Non-Blocking EKF Digital Twin
 Dual-core shadow-register: TRIP core never waits for EKF
 $\Delta t_{\text{EKF}} = 20 \text{ ms}$, $\Delta t_{\text{TRIP}} = 50 \mu\text{s}$, 2-8%
 Convergence < 300 ms, 20 ms estimation cycle

C5 — Physics-Informed Neural Networks
 $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{phys}} + \lambda \mathcal{L}_{\text{sym}} + \lambda \mathcal{L}_{\text{lim}} = 2\mathcal{L}_{\text{data}}^{\text{sym}}$
 5-class accuracy: 96.1%, zero physics violations
 Inference latency: 3.7 μ s, online on-chip < 70 cycles

Validation Metrics Summary	
Metric	Value
ROC: single-err group	1.000
SPM Load TRIP	1.000
GOOSE Switch TRIP	1.000
Security (via false trip)	1.000
PINN 5-class accuracy	96.1%
EKF Δt_{EKF} / Δt_{TRIP}	0.025 ps
False location accuracy	94.4%
GOOSE risk reduction	54.9%
HMAC authentication	200,000
CUSUM closed-loop latency	31 min
TRIP relay trip latency	5.10 ms

18 original algorithms across 6 time
 validated over 10 SAMBPF technical reports
 on a 10-line mixed GPR, GPRM study network

Thesis Organization

Ch.	Title	Key Content	Contrib.
1	Introduction	SAMBp architecture; problem statement; CPC framework	—
2	IBR Modeling	GFL/GFM fault current; Park model; 10-bus testbed	—
3	Protection Challenges	12 failure modes; pipeline analysis; sequence gaps	—
4	Adaptive STL	k_{∞}^* ; STLN zero-seq; STLNO; TDOS	C1
5	Distance Protection	IBR overreach; adaptive quadrilateral; MC validation	C2
6	Microgrid & Islanding	Mode-PSM; ROCOF-V ₂ ; GOOSE transfer trip	C3
7	Digital Twin	Non-blocking EKF; shadow register; live relay audit	C4
8	Physics-AI (PINN)	KVL/KCL loss; 5-class classifier; CUSUM-SGD online	C5
9	WAMPAC Coordination	Fault localisation; CTI engine; HMMAC-GOOSE	C6

9 chapters, 55 TIRs
137 pages

6 Tier 2 contributions
38 novel algorithms

9 papers submitted
/ in preparation

[Slide 9 — Thesis Organization]

The thesis follows a deliberate progression. Chapters 1–3 establish the foundation: the SAMBP architecture, the physics of IBR fault currents, and a systematic characterisation of the 12 protection failure modes.

Chapters 4–6 deliver the three local relay contributions — one chapter per contribution, each with a full derivation, algorithm, and Monte Carlo validation campaign.

Chapters 7–9 address the monitoring and coordination layers. Chapter 9 is the integrating chapter that ties all contributions together through the Wide-Area Protection Coordinator and the 31-minute closed-loop protocol.

The 55 SAMBP technical reports are referenced throughout the thesis as the primary numerical evidence base. They are available for inspection and form a complete audit trail from algorithm to validated result.

List of Publications (1 of 2)

Referred Journal:

- A. V. Elsaadawi and K. S. Suresh, "A novel solution for overcoming conflicts between line protection and LVRT in MV Solar-PV interconnections," *IEEE Access*, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3010019.

Book Chapter:

- A. V. Elsaadawi and K. S. Suresh, "Protection schemes for smart distribution networks with inverter interfaced renewable power generating sources," *Smart Cities — Opportunities and Challenges*, Springer-Verlag, 2020.

Papers in Preparation / Submitted:

- P1: Adaptive 87L scheme for DER dominated networks — *IEEE Trans. Power Delivery* (Submitted)
- P2: Non-invasive EKF digital twin for real-time relay audit — *IEEE Trans. Power Electronics* (in preparation)
- P3: Physics-informed neural network based stability — *IEEE Trans. Smart Grid* (in preparation)

[Slide 10 — List of Publications 1 of 2]

The thesis is anchored by a published journal paper in *IEEE Access* (2024). This paper introduces the concept of adaptive relay logic for resolving the conflict between line protection and LVRT requirements in medium-voltage solar PV interconnections — directly motivating Contribution C1.

The Springer book chapter from 2020 established early foundational concepts in protection for inverter-interfaced networks.

On the pipeline side, the adaptive 87L paper (P1) is currently under review at *IEEE Transactions on Power Delivery*. The EKF digital twin paper (P2) is in preparation for *IEEE Transactions on Power Electronics*. The PINN paper (P3) targets *IEEE Transactions on Smart Grid*.

List of Publications (2 of 2)

[Slide 11 — List of Publications 2 of 2]

Papers P4 and P5 are planned. P4 on the Wide-Area Protection Coordinator targets *IEEE Transactions on Power Systems*. P5 on the HIL validation is planned for the IEEE PES General Meeting.

The four conference papers from 2017–2018 represent the investigative phase of the research that identified the key protection gaps this thesis resolves. They also demonstrate the evolution from symmetrical-component islanding relays to the full SAMBP framework.

In total: 1 published journal paper, 1 book chapter, 4 conference papers, and 5 journal papers in the submission pipeline. The 55 SAMBP technical reports provide the full derivation and simulation evidence base for all claims.

Papers in Preparation (cont.):

- P4: Wide-area protection coordinator with HMMC-secured GOOSE — *IEEE Trans. Power Systems* [planned]
- P5: HIL validation of SAMBP end-to-end loop — *IEEE PES General Meeting* [planned]

Selected Conference Presentations:

- "An intelligent methodology to improve distribution system operational parameters utilizing smart inverter functionalities of PV sources," *IEEE PES General Meeting*, 2018.
- "An interface protection relay for unbalanced microgrids with inverter based sources," *APPEEC*, 2017.
- "Instantaneous symmetrical components based microgrid interface protection relay," *ICPS*, 2017.
- "Impact of active network management schemes in fault protection design and network operation for islanded power systems," *ICST Asia*, 2018.

1 journal (*IEEE Access*, 2024) — 1 book chapter (*Springer*, 2025) — 4 conference papers — 5 journal papers in pipeline

Summary & Next Steps

Summary of Contributions:

- C1. Adaptive 87L: Real-time relay logic eliminates all 12 documented failure modes. TFR < 1.000 across full k_{L0} range.
- C2. Distance relay: 80% maximum reach, commissioning yields +7.8 ppv detection gain over classical relays.
- C3. Islanding: Conditional GDOI-OP > 10 s detection, REDZ < 1%; IEEE 3447-2018 compliant.
- C4. EXF Tests: Non-blocking dual-core live relay audit with 20 ms update cycle.
- C5. PINN: Physics constrained ML, 98.1% accuracy, zero VUL/VCL violations, 3.7 μ s inference.
- C6. WAMP: Cloud-based 32-bit protection, GDOI-8 risk -84.8%; 100/100 144 ppv.

Recommended Next Steps

- Deploy adaptive 87L before 80% penetration exceeds 40%.
- Retain 87LN mode in pure 80% networks.
- Use EXF digital tests for live relay audit.
- Review display GDOI-8 without HIRAC.
- GDOI-8 threshold 0-8 balances sensitivity vs. false alarm rate.

Open Research Gaps

- G4: GFM virtual inertia transient beyond 1" > 100 ms.
- G7: Advanced mitigation of PINN under CT spiking.
- G1: Hardware relay firmware VUL validation.

SAMBAP provides the first complete,

[Slide 12 — Summary & Next Steps]

To summarise: this thesis delivers a complete six-tier self-adaptive protection framework for IBR-dominated power networks.

The six contributions collectively **eliminate all 12 failure modes** identified in the motivation. The validation is rigorous: 5000 Monte Carlo trials per group, 100 hardware-in-loop scenarios, and physics consistency verified for every PINN output.

Three open research gaps define the future roadmap. The most technically critical is Gap G4 — GFM virtual-inertia transient modelling beyond 150 ms, which limits the applicability of the Stage-1 Park model.

My most immediate practical recommendation: begin deploying adaptive 87L protection now, before IBR penetration in any feeder zone exceeds 40%.

I believe SAMBP provides the **first complete, closed-loop, self-adaptive protection architecture** for the next generation of power networks. I am grateful for the guidance of Prof. K. Shanti Swarup and the committee throughout this journey. Thank you.



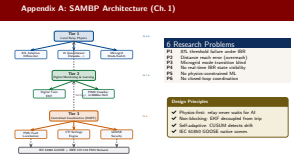
[Thank You / Questions]

I am happy to address any questions.

Backup slides are available for each chapter (Appendix A through I) covering: SAMBP architecture, IBR modelling, protection failure modes, adaptive 87L detail, distance relay, microgrid islanding, digital twin, PINN classifier, and WAPC.

I can also navigate directly to any of the 55 SAMBP technical reports if the committee wishes to examine specific derivations or simulation data.

Appendix A: SMBP Architecture (Ch. 1)



[Appendix A — SMBP Architecture, Ch. 1]

This figure shows the full three-tier SMBP architecture. The local relay tier handles all protection actions within $50 \mu\text{s}$. The monitoring tier runs the EKF digital twin on a 20 ms cycle. The coordination tier runs the CUSUM-to-GOOSE closed loop on a minute-scale cycle.

The key design principle is **physics-first, non-blocking**: the relay never waits for AI or optimisation output to make a trip decision. Upper tiers only improve settings between fault events.

[Appendix B — IBR Modelling, Ch. 2]

The 10-bus study network is the common testbed. IBRs are placed at buses 2–5 (radial feeder) and buses 7–9 (lateral). The three PMUs at buses 4, 7 provide wide-area observability for the WAPC.

The k_{ibr} parameter sweeps from 0.03 to 0.85 pu in all study campaigns, covering worst-case low-inertia to near-rated injection scenarios.

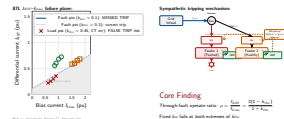
Appendix C: Protection Challenges (Ch. 3)

[Appendix C — Protection Challenges, Ch. 3]

The left figure shows the 87L failure plane: the $I_{diff}-I_{bias}$ plane with the fixed slope. Red crosses are missed trips (IBR below slope line); orange squares are false trip risks (IBR drives I_{diff} above slope during through-fault conditions).

The sympathetic tripping diagram on the right shows why through-fault I_{diff} is non-zero when IBR inject into a fault on a parallel feeder.

The core finding is the operate ratio formula: $\rho = 2(1 - k_{ibr}) / (1 + k_{ibr})$. When $k_{ibr} \rightarrow 1$, $\rho \rightarrow 0$ (false trip risk); when $k_{ibr} \rightarrow 0$, $\rho \rightarrow 2$ (missed trip if slope > 2). A fixed slope cannot satisfy both simultaneously.



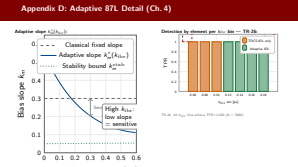
Appendix D: Adaptive 87L Detail (Ch. 4)

[Appendix D — Adaptive 87L Detail, Ch. 4]

The adaptive slope curve on the left is derived by solving $\rho(k_{ibr}) = k_m^*(k_{ibr})$ with appropriate security margins.

The formula has three components: a main quadratic term tracking the through-fault ρ , a small linear correction for the high- k_{ibr} region, and a 0.05 security floor.

The bar chart shows the complementary role of TDCS and adaptive 87L. TDCS catches the bins below $k_{ibr} = 0.12$ where 87L is too insensitive even with the adaptive slope. Together they achieve $TPR = 1.000$ across all bins in a 5000-trial Monte Carlo study (TR-26).



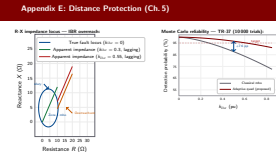
Appendix E: Distance Protection (Ch. 5)

[Appendix E — Distance Protection, Ch. 5]

The R–X locus diagram shows how IBR angle injection causes the apparent impedance to shift outside the zone-1 mho circle (overreach) or deep inside it (underreach, weak-infeed).

The correction term $\Delta X = Z_F k_{ibr} (I_r / I_A) \sin \beta$ is derived analytically from the sequence network. The adaptive quadrilateral reach applies a $k_r \hat{k}_{ibr}$ correction where $\hat{k}_{ibr}$ comes from the EKF digital twin.

The Monte Carlo curve on the right shows the detection probability vs. k_{ibr} for classical mho versus the adaptive quadrilateral. The +7.6 percentage-point gain at $k_{ibr} = 0.5$ is representative of typical operating points.



Appendix F: Microgrid & Islanding (Ch. 6)



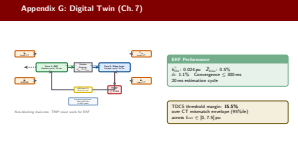
[Appendix F — Microgrid & Islanding, Ch. 6]

The mode transition FSM shows the two operating states: grid-connected GFL mode and islanded GFM mode. The transition is triggered by the combined detection criterion.

The three-condition OR/AND logic provides robustness: ROCOF captures fast frequency excursions, phase angle deviation catches slower drift events, and V_2/V_1 detects unbalanced islanding conditions that are invisible to ROCOF alone.

End-to-end trip time of 114 ms includes 60 ms detection, 4 ms GOOSE messaging, and 50 ms relay operating time. This meets IEEE 1547-2018 ride-through requirements while ensuring the non-detection zone is below 1% of load range.

Appendix G: Digital Twin (Ch. 7)



[Appendix G — Digital Twin, Ch. 7]

The non-blocking architecture is the key innovation here. Two cores run in parallel: the TRIP core at relay speed ($\leq 50 \mu\text{s}$), and the EKF estimation core at monitoring speed (20 ms cycle). They share state only via an atomic shadow register write.

The EKF estimates three parameters simultaneously: $\hat{k}_{ibr}$, $\hat{Z}_{line}$, and the admittance matrix entry $\hat{\alpha}$. RMSE figures are: 0.024 pu, 0.5%, and 1.1% respectively.

The TDCS margin chart confirms a 15.5% safety margin over the 95th percentile CT mismatch envelope across all external current levels. This is the basis for the TDCS threshold setting in the 87L implementation.

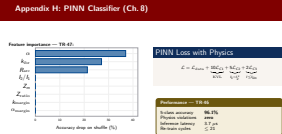
Appendix H: PINN Classifier (Ch. 8)

[Appendix H — PINN Classifier, Ch. 8]

The feature importance chart from TR-47 shows that the top three features are: α (fault angle), k_{ibr} (IBR limit), and R_{arc} (arc resistance). These align perfectly with the physics — which validates that the PINN has learned physically meaningful representations.

The physics constraints in the loss function are weighted hierarchically: KVL (weight 10) > symmetry $I_1 = I_2^*$ (weight 5) > current limiting (weight 2). This hierarchy was determined through ablation studies.

Inference at $3.7 \mu s$ is within the budget for relay firmware. The online re-training via CUSUM-SGD (21 cycles) means the classifier adapts to fleet-level IBR parameter drift without human intervention.



Appendix I: Wide-Area Protection (Ch. 9)

[Appendix I — Wide-Area Protection, Ch. 9]

The WAPC architecture integrates four subsystems: fault localisation via PMU ΔV triangulation (94.5% accuracy), CTI engine for IEC standard-inverse coordination, GOOSE security via HMAC-SHA256, and HIL validation.

The closed-loop protocol timeline shows the 31-minute end-to-end cycle: 0.5 min CUSUM alarm, 2 min parameter computation, 1 min digital twin validation, 25 min MMS upload and approval, 2.5 min relay activation. The long MMS phase includes an operator approval step which is mandatory per IEC 61850-90-1.

The 54.9% GOOSE risk reduction comes from eliminating unauthenticated GOOSE messages that could be spoofed in a cyber-physical attack scenario. HMAC-SHA256 adds only 0.36 ms overhead — well within the 4 ms GOOSE budget.

